Zhang 2021 RCC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/8 samples (≥369 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→3.17 of 7); clear-cell and chromophobe RCC intermix. Both compartments (Cycling/Metabolic 9,545 / Embryonic 1,673)

Gate 2 — module geography → GO (non-branching)

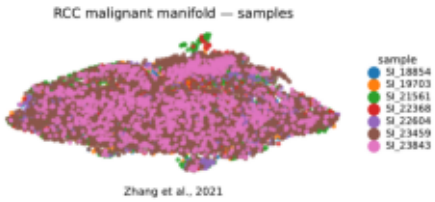
Single dominant pSom→Crest axis: pSom high over the body declines along pseudotime (ρ=−0.27); Crest anti-correlated (r=−0.91), rising to a terminal readout (ρ=+0.48; crest-high decile mean pt=0.72). Gast weakly structured, co-varies with pSom (early, spent).

Gate 3 — hub mutual exclusivity → NA

Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

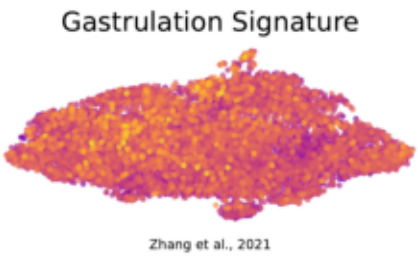
Gate 1: integration



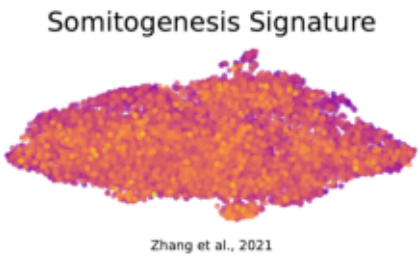
Trajectory / root



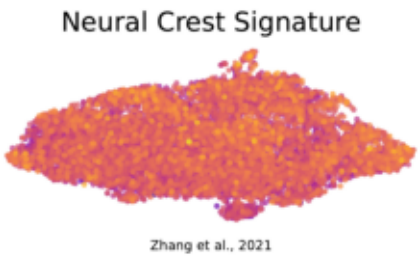
Gast



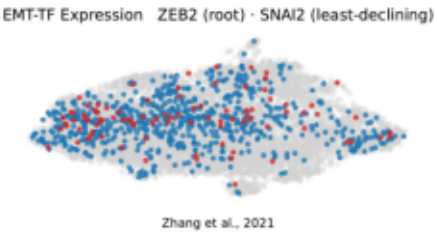
pSom



Crest



EMT-TFs



Dataset. Renal Cell Carcinoma (Zhang et al., 2021), 10x — 8 malignant samples / 11,410 malignant cells; 11,218 across 7 samples (≥369) enter preprocessing (0 island cells removed); the whole malignant manifold (no lobe scoping) enters the trajectory. Mixed clear-cell + chromophobe RCC.

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zhang's RCC malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both proliferation compartments (module–g2m |ρ|≤0.03) and both RCC subtypes carry near-identical developmental means, so the whole manifold is kept (Bischoff/Sun no-scoping precedent). A curve is fold-free (self-approach 2.76); a PPT tree's forks do not resolve a Gast/pSom split (over-fitting of a near-round cloud). EMT-TFs: ZEB2 is the root-end factor (most negative ρ); no well-expressed factor is genuinely terminus-enriched (all ρ≤0), so SNAI2 is only the least-declining — all EMT-TFs sit near the 10x detection floor.

Parameter choices

Min cells/sample	≥369 (7/8; gap 710→192, ratio 3.7)	**Integration**	Harmony θ=10; PCA50; n_pcs=20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100